

# class 5: Data Viz w/ ggplot2

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### ##Background

There are many graphics system in R for making plots and figures. These include so-called “*base R*” *graphics* like the ‘plot()’ function and add on packages like **ggplot2**.

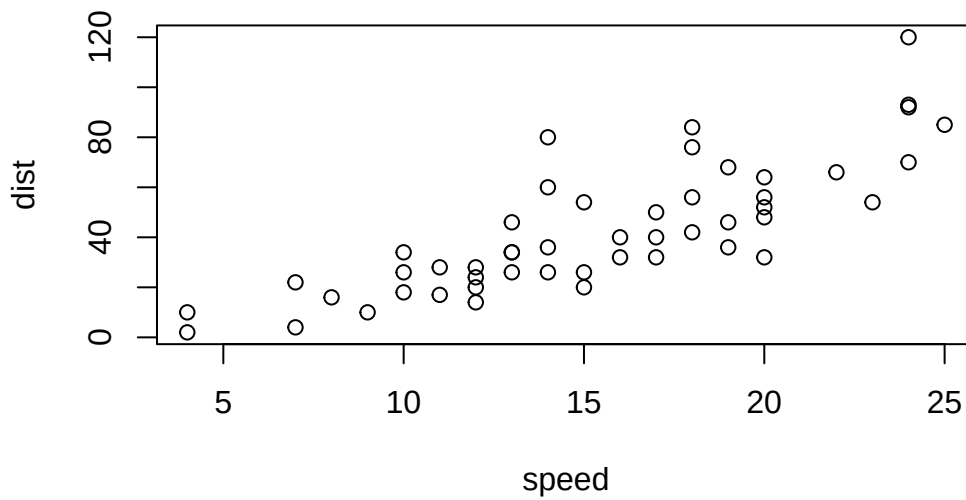
Let’s compare how we make a simple figure with these two systems:

We can use the in-built “cars” dataset:

```
head(cars)
```

```
      speed dist
1         4    2
2         4   10
3         7    4
4         7   22
5         8   16
6         9   10
```

```
plot(cars)
```



Before I can use `ggplot2` I need to install it on my computer. To do this we can use the function `install.packages(ggplot2)`

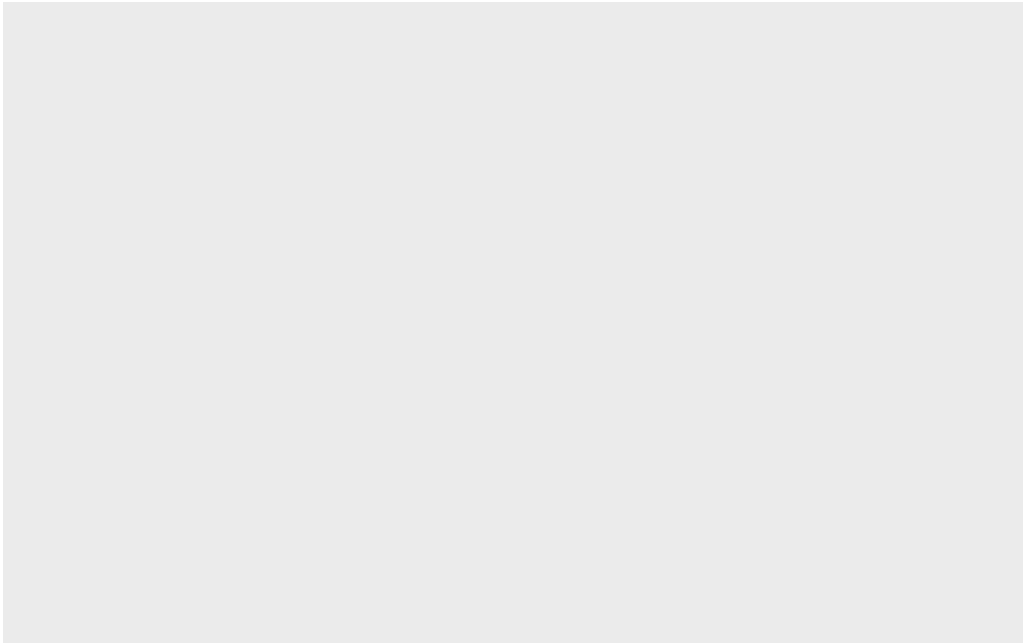
**N.B.** We never run `install.packages()` in our quarto doc (we run it once only in our R console) as it would re-install the package every time we render our quarto package

Once installed we need to load up the package into our R brain:

```
library(ggplot2)
```

The main function in the **ggplot2** package is called `ggplot()`

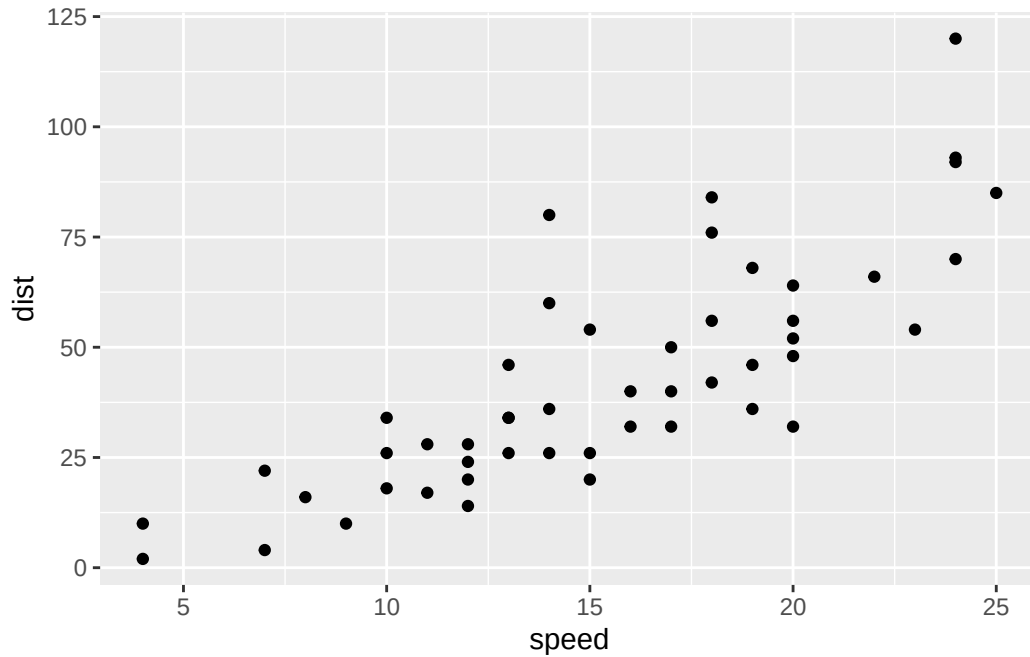
```
ggplot(cars)
```



every ggplot has at least 3 layers:

- the **data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps on the plot),
- the **geom** layer (how you want the plot drawn, e.g points, lines, etc.)

```
ggplot(cars)+  
  aes(x= speed, y= dist) +  
  geom_point()
```

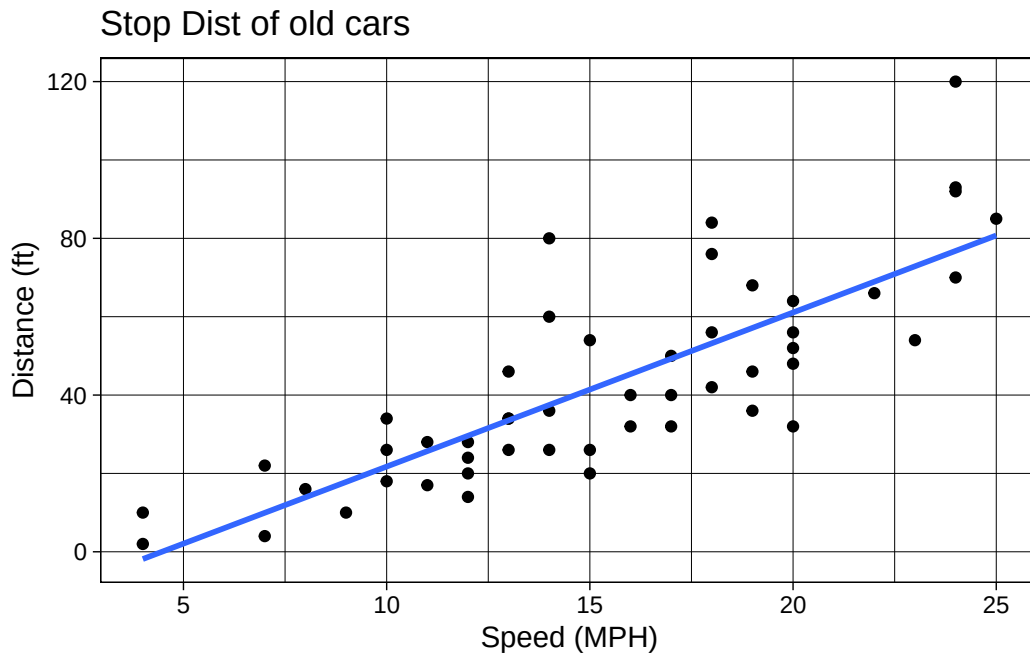


### Add some custom features

Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars)+  
  aes(x= speed, y= dist) +  
  geom_point() +  
  geom_smooth(method = "lm", se= FALSE) +  
  theme_linedraw() +  
  labs(title="Stop Dist of old cars",  
        x="Speed (MPH)",  
        y= "Distance (ft)")
```

``geom_smooth()`` using formula = 'y ~ x'



Q. Can you make the 'geom\_smooth()' function produce a linear straight line fit to the data and turn off the "gray" error region.

## Gene Expression

Import the data to plot

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

|   | Gene       | Condition1 | Condition2 | State      |
|---|------------|------------|------------|------------|
| 1 | A4GNT      | -3.6808610 | -3.4401355 | unchanging |
| 2 | AAAS       | 4.5479580  | 4.3864126  | unchanging |
| 3 | AASDH      | 3.7190695  | 3.4787276  | unchanging |
| 4 | AATF       | 5.0784720  | 5.0151916  | unchanging |
| 5 | AATK       | 0.4711421  | 0.5598642  | unchanging |
| 6 | AB015752.4 | -3.6808610 | -3.5921390 | unchanging |

```
sum(genes$State == "up")
```

```
[1] 127
```

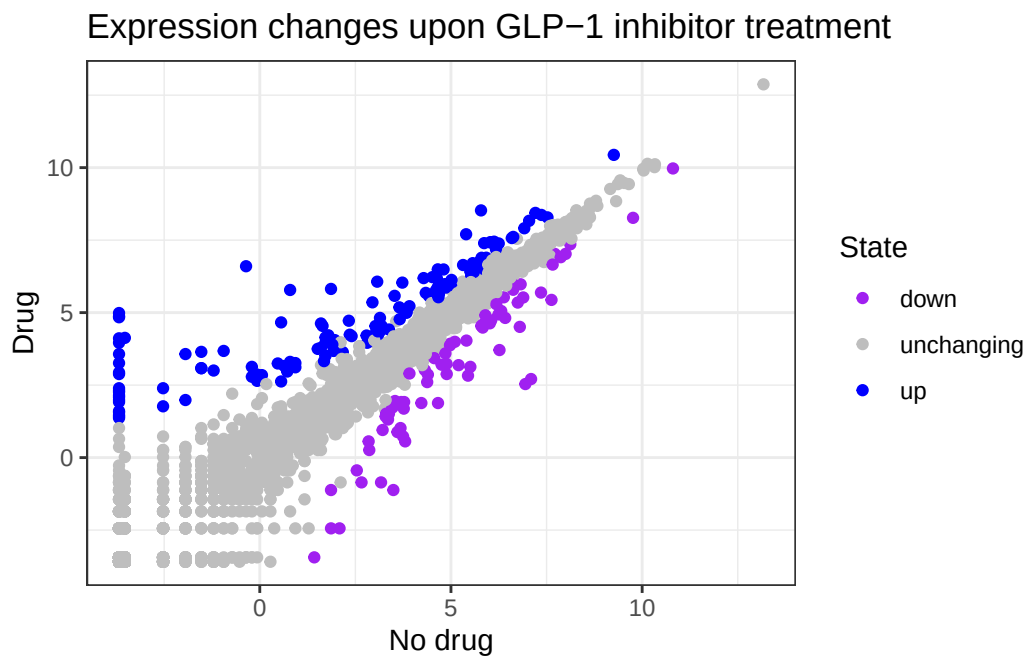
A useful new function in this context is the 'table()' function:

```
table(genes$State)
```

| down | unchanging | up  |
|------|------------|-----|
| 72   | 4997       | 127 |

My first plot attempt

```
ggplot(genes) +  
  aes(Condition1, Condition2, col=State) +  
  geom_point() +  
  scale_color_manual(values=c("purple", "gray", "blue")) +  
  theme_bw() +  
  labs(x="No drug", y="Drug",  
       title="Expression changes upon GLP-1 inhibitor treatment")
```



## Going Further

Here we read the famous gapminder dataset:

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)
```

|   | country     | continent | year | lifeExp | pop      | gdpPercap |
|---|-------------|-----------|------|---------|----------|-----------|
| 1 | Afghanistan | Asia      | 1952 | 28.801  | 8425333  | 779.4453  |
| 2 | Afghanistan | Asia      | 1957 | 30.332  | 9240934  | 820.8530  |
| 3 | Afghanistan | Asia      | 1962 | 31.997  | 10267083 | 853.1007  |
| 4 | Afghanistan | Asia      | 1967 | 34.020  | 11537966 | 836.1971  |
| 5 | Afghanistan | Asia      | 1972 | 36.088  | 13079460 | 739.9811  |
| 6 | Afghanistan | Asia      | 1977 | 38.438  | 14880372 | 786.1134  |

Q. How many entries (i.e. rows) are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many different country entries are in this dataset?

```
length(table(gapminder$country))
```

```
[1] 142
```

```
length(unique(gapminder$country))
```

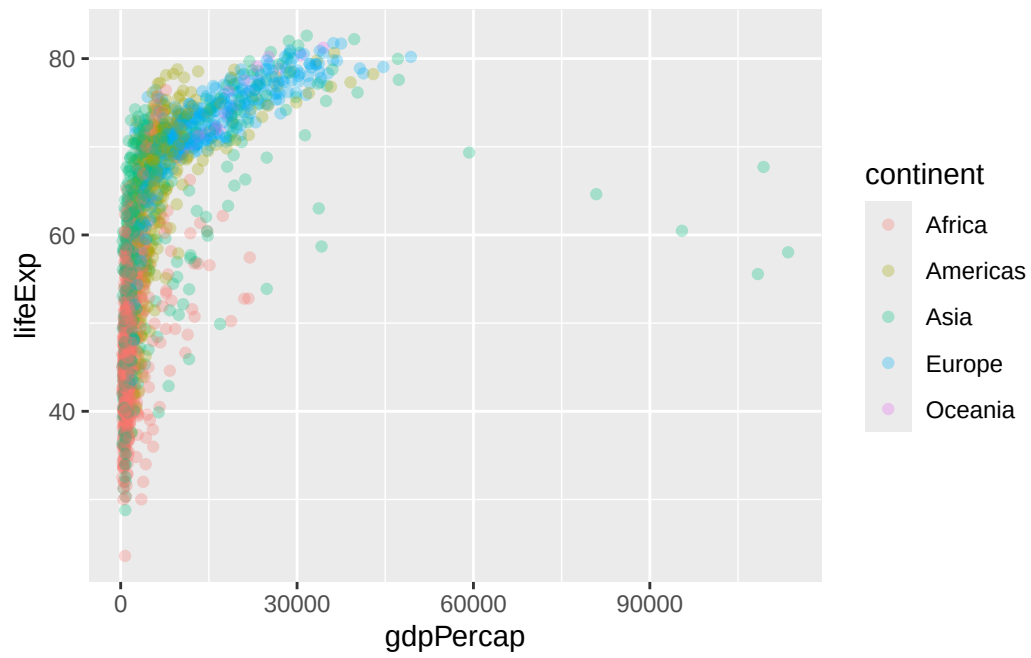
```
[1] 142
```

Let's make our first plot of the entire dataset

Plot of “gdppPercap” vs “lifeExp” colored by “continent”

```
p <- ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col= continent) +
  geom_point(alpha= 0.3)
```

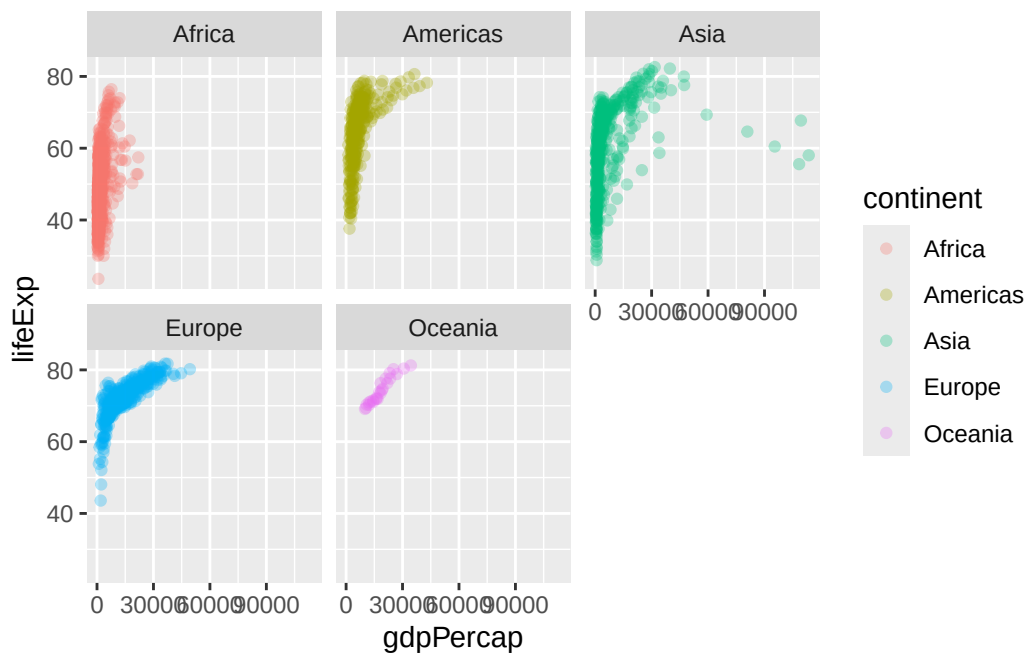
p



I can add more layers to p ...

```
p +  
  facet_wrap(~continent)
```





make a plot for years 1977 and 2007 only.

Q. First use the **dplyr** package and the 'filter()' function from that package to extract the rows from the year 2007.

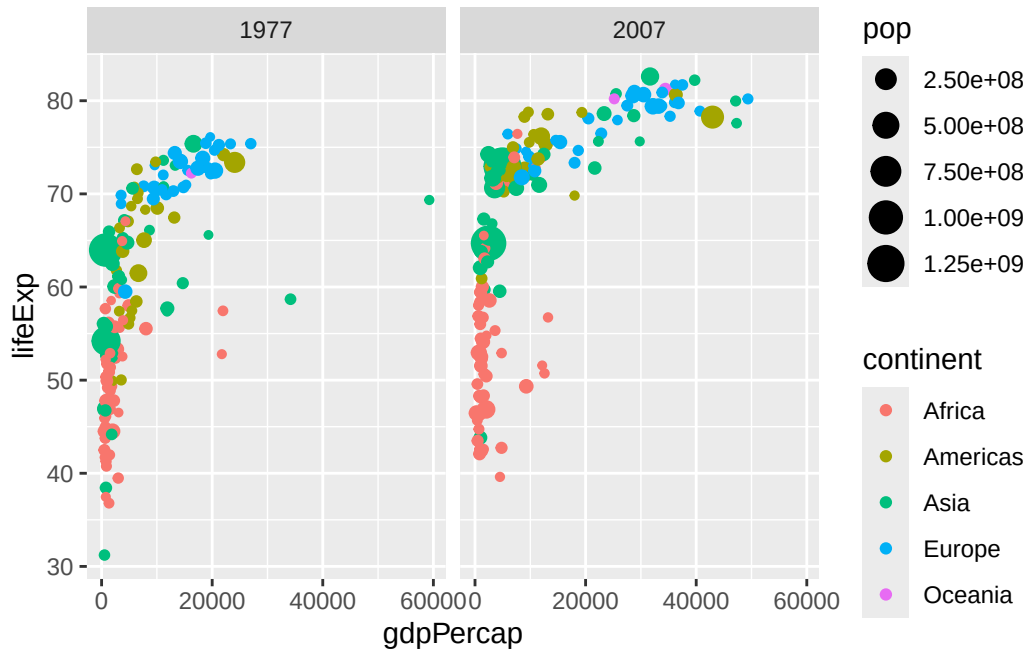
```
library(dplyr)
```

```
head(filter(gapminder, year == 2007))
```

|   | country     | continent | year | lifeExp | pop      | gdpPercap  |
|---|-------------|-----------|------|---------|----------|------------|
| 1 | Afghanistan | Asia      | 2007 | 43.828  | 31889923 | 974.5803   |
| 2 | Albania     | Europe    | 2007 | 76.423  | 3600523  | 5937.0295  |
| 3 | Algeria     | Africa    | 2007 | 72.301  | 33333216 | 6223.3675  |
| 4 | Angola      | Africa    | 2007 | 42.731  | 12420476 | 4797.2313  |
| 5 | Argentina   | Americas  | 2007 | 75.320  | 40301927 | 12779.3796 |
| 6 | Australia   | Oceania   | 2007 | 81.235  | 20434176 | 34435.3674 |

```
g07 <- filter(gapminder, year == 2007)
g77 <- filter(gapminder, year == 1977)
g <- filter(gapminder, year == 2007 | year == 1977)
```

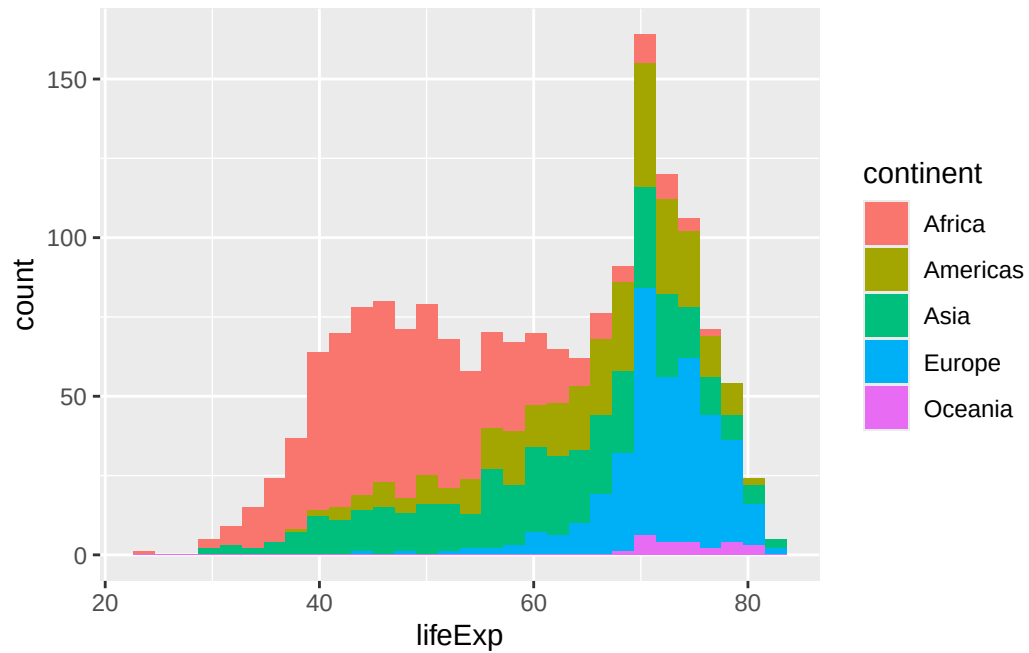
```
ggplot(g) +
  aes(gdpPercap, lifeExp, col=continent, size=pop) +
  geom_point() +
  facet_wrap(~year)
```



Q. Make a histogram of life exp colored by continent Q. Make a histogram of lifeExp faceted by continent

```
ggplot(gapminder) +
  aes(x = lifeExp, fill = continent) +
  geom_histogram()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
ggplot(gapminder) +  
  aes(lifeExp, fill = continent) +  
  geom_histogram() +  
  facet_wrap(~ continent)
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

